

# Swap space

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.2 Memory and storage — Paper 1, Computer systems                                                       |
| Lesson          | Swap space (lesson 2 of the unit)                                                                                 |
| Lesson ID       | ms-102                                                                                                            |
| Experience ID   | ms-102                                                                                                            |
| Plan ID         | PLAN-MS-L02                                                                                                       |
| Content version | 8b28425dd472                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=ms-102">https://revise360.co.uk/experience.html?id=ms-102</a> |

## Big question, outcomes and success criteria

**Big question** — Why do we need virtual memory, and what can we do to avoid using it?

### Prerequisites

The lesson before this one is ms-101. Pupils should already have finished it: the starter below retrieves from it.

### Assessable outcomes

1. **ms-102-o1** — understand the need for virtual memory
2. **ms-102-o2** — understand how virtual memory works

## Vocabulary, resources and preparation

### Key vocabulary — Virtual memory, Volatile

### Resources

|                   |                                            |
|-------------------|--------------------------------------------|
| SLIDES            | presentations/MS_L02_SwapSpace_Lesson.pptx |
| WORKSHEET         | worksheets/MS_L02_SwapSpace_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=ms-102                  |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build APP\_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on MS_L02_SwapSpace_Worksheet.docx is filled in.                       |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find When RAM runs out.                               | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. When RAM runs out

- Every open program needs space in RAM.
- When RAM is full, something has to move out to make room.
- Without virtual memory, the computer would refuse to open anything else.

**Challenge on this wall:** What happens if you open a graphics app when RAM is already full?

### 2. What is virtual memory?

- A section of secondary storage used as if it were extra RAM.
- Data not currently needed is moved there, freeing RAM for what is needed.
- The CPU can only work with data in RAM, so anything needed must be moved back first.

**Challenge on this wall:** Write a definition of virtual memory.

### 3. Swapping

- Data moves out of RAM to disk, and back again when needed: this is swapping, or paging.
- If the CPU needs something that isn't in RAM, something else must move out to make room.
- Every swap takes time, because secondary storage is slow.

**Challenge on this wall:** Explain what happens when a program in virtual memory is needed again.

### 4. Why it's slow

- Secondary storage is thousands of times slower than RAM.
- Heavy swapping is sometimes called disk thrashing: the computer spends more time swapping than working.
- You notice it as constant disk activity and everything feeling sluggish.

**Challenge on this wall:** Why does a computer relying on virtual memory feel so slow?

### 5. Avoiding virtual memory

- Install more RAM so more can stay in memory.
- Close programs you aren't using.
- An SSD makes swapping less painful, but it's still far slower than RAM.

**Challenge on this wall:** Why does doubling the RAM make a computer significantly faster?

### 6. Virtual memory in action

- A user has a word processor and a browser open, filling RAM.
- They open a graphics app, so the least-used data is swapped out to disk.
- Switching back to the word processor swaps it in again, which is why the switch feels slow.

**Challenge on this wall:** Illustrate this scenario on your worksheet.

## Retrieval and the check for understanding

**Starter** — You open ten browser tabs, a game and a video editor on an old laptop. Describe what happens and why.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ms-102-s4-c2), so the wording a pupil sees matches the wording you say.

#### What is disk thrashing?

**Correct:** The computer spending most of its time swapping data instead of doing useful work

#### What a pupil might choose instead

- A hard disk being physically damaged
- Deleting temporary files
- Defragmenting the disk

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** It's the point where virtual memory stops helping.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

| STAGE       | ACTIVITIES | WHAT IT IS                                                                                                        |
|-------------|------------|-------------------------------------------------------------------------------------------------------------------|
| Guided      | 12         | Support is in the activity: an example to run, an output to predict, one thing to change, a gap to fill.          |
| Independent | 0          | The pupil writes or fixes it with the brief and nothing else.                                                     |
| Challenge   | 0          | The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station. |

Every activity in this lesson is guided: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| # | STATION                  | ACTIVITIES | RECORD ON PAPER                           |
|---|--------------------------|------------|-------------------------------------------|
| 1 | When RAM runs out        | 2          | When RAM is full...                       |
| 2 | What is virtual memory?  | 2          | Virtual memory is...                      |
| 3 | Swapping                 | 1          | Swapping means...                         |
| 4 | Why it's slow            | 2          | Virtual memory is slow because...         |
| 5 | Avoiding virtual memory  | 2          | To avoid using virtual memory, you can... |
| 6 | Virtual memory in action | 2          | In this scenario...                       |

| #            | STATION                          | ACTIVITIES | RECORD ON PAPER                       |
|--------------|----------------------------------|------------|---------------------------------------|
| <b>final</b> | <b>Final challenge: the swap</b> | <b>1</b>   | the answer to the station's challenge |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                       | MARKS    | WHERE                                                  |
|-----------|----------------------------------------------------------------|----------|--------------------------------------------------------|
| ms-102-x1 | Describe what is meant by virtual memory.                      | <b>2</b> | worksheet, Exam practice a); deck, Exam practice slide |
| ms-102-x2 | Explain why using virtual memory slows a computer down.        | <b>3</b> | worksheet, Exam practice b); deck, Exam practice slide |
| ms-102-x3 | State one way a user could reduce the need for virtual memory. | <b>1</b> | worksheet, Exam practice c); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ms-102.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What happens when RAM becomes full?** — The computer switches off; The CPU slows its clock
- **Which is most likely to fill RAM?** — Having a large hard drive; Having a fast CPU
- **What is virtual memory?** — Memory that only exists in the cloud; A faster type of RAM
- **Can the CPU work directly on data held in virtual memory?** — Yes, just as quickly as RAM; Yes, but only for small files
- **Why is virtual memory slower than RAM?** — It uses a different power supply; It is stored in the cloud

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- What virtual memory is
- How data is swapped between RAM and disk
- Why virtual memory is slow

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is ms-103, Storage warehouse. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

## Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

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### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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